Fizika yechimlari

1. Sistemaning og'irlik markazi *2m* massali sharchadan qanday masofada yotadi (1–rasm *a*) ?

Berilgan: $m,\ a$;

Topish kerak: $d-?$

Masalaning yechilishi:

Koordinata boshini *m* massali jism bilan bog'lab, *Ox* o'qini gorizontal, *Oy* o'qini unga tik yo'naltiramiz (1–rasm *b*). U holda jismlarning mazkur o'qlar bo'yicha koordinatalari mos ravishda quyidagicha bo'ladi:

$$x_1=0,\ y_1=0;\ x_2=a,\ y_2=0;\ x_3=\frac{a}{2},\ y_3=\frac{\sqrt{3}}{2}a;$$

Jism koordinatalarining bu qiymatlarini sistemaning massa markazini aniqlash formulasi

$$x_0=\frac{m_1x_1+m_2x_2+m_3x_3+...+m_nx_n}{m_1+m_2+m_3+...+m_n}$$

ga qo'yamiz hamda hisoblash ishlarini bajaramiz:

$$x_0=\frac{m\cdot 0+m\cdot a+2m\cdot\frac{a}{2}}{m+m+2m}=\frac{a}{2},\qquad y_0=\frac{m\cdot 0+m\cdot 0+2ma\cdot\frac{\sqrt{3}}{2}}{m+m+2m}=\frac{a\sqrt{3}}{4}$$

2m massali sharchadan sistemaning og'irlik markazigacha bo'lgan *d* masofa 1–rasm (*b*) dan, ikki nuqta orasidagi masofani topish tenglamasidan kelib chiqadi:

$$\boxed{d=\frac{a\sqrt{3}}{4}}$$

2. Tomonlari 20 sm dan bo'lgan teng tomonli uchburchak uchlarida va markazida joylashgan, massalari $m_1=100\ g, m_2=200\ g, m_3=300\ g$ va $m_4=400\ g$ bo'lgan to'rtta sharchadan iborat sistema massalar markazining koordinatalarini toping. Koordinata o'qlarining yo'nalishini 2–rasmdagidek tanlab oling.

Berilgan: $a = 0{,}2\ m,\ m_1 = 0{,}1\ kg,\ m_2 = 0{,}2\ kg,\ m_3 = 0{,}3\ kg,\ m_4 = 0{,}4\ kg;$

Topish kerak: $x - ?\quad y - ?$

Masalaning yechilishi:

Koordinata boshini *2* sharcha bilan bog'lab, har bir sharni *Ox* hamda *Oy* o'qlari bo'yicha koordinatalarini mos ravishda quyidagicha yozish mumkin:

$$x_1 = \frac{a}{2},\ y_1 = \frac{a\sqrt{3}}{6};\quad x_2 = 0,\ y_2 = 0;\ x_3 = \frac{a}{2},\ y_3 = \frac{a\sqrt{3}}{2};\quad x_4 = a,\ y_4 = 0;$$

Sharchalarning koordinatalarini sistemani massa markazini aniqlash formulasiga qo'yamiz hamda hisoblash ishlarini bajaramiz:

$$x_0 = \frac{m_1x_1 + m_2x_2 + m_3x_3 + m_4x_4}{m_1 + m_2 + m_3 + m_4} = 0{,}12\ \ m;$$

$$y_0 = \frac{m_1y_1 + m_2y_2 + m_3y_3 + m_4y_4}{m_1 + m_2 + m_3 + m_4} = 0{,}058\ m;$$

Hisoblash natijalari: $x_0 = 0{,}12\ m,\ y_0 = 0{,}58\ m.$

3. Fanerdan qirqib olingan, tomoni 20 sm bo'lgan kvadratdan hamda tomoni 30 sm bo'lgan teng tomonli uchburchakdan iborat uychaning og'irlik markazini aniqlang (3–rasm).

Berilgan: $a = 0{,}20\ m,\ b = 0{,}30\ m;$

Topish kerak: $y_0 - ?$

Masalaning yechilishi:

Kvadratning og'irlik markazi uning geometrik markazi 0_1 nuqtada, uchburchakning og'irlik markazi esa uning medianalari kesishish nuqtasi 0_2 nuqtada bo'ladi. U holda «uycha» ning og'irlik markazi 0_1 va 0_2 nuqtalar orasida bo'ladi.

Sistemaning og'irlik markazini aniqlash ifodasidan foydalanib, *O* nuqtaga nisbatan quyidagi tenglamani yozamiz:

$$y_0 = \frac{(a/2)m_1 + (a + b/2\sqrt{3})m_2}{m_1 + m_2} \qquad (1)$$

Bu yerda m_1 –kvadratning massasi, m_2 – uchburchakning massasi. $m = \rho V$ ifodaga asosan ular uchun quyidagi tengliklarni yozamiz:

$$m_1 = \rho d a^2, \quad m_2 = \rho d b^2 \sqrt{3}/4$$

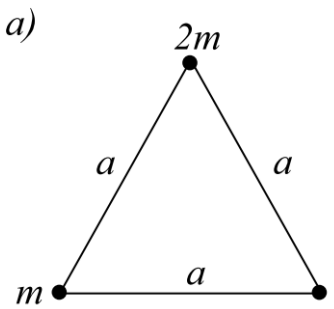
Bu yerda ρ –faren materialining zichligi, d–uning qalinligi. Oxirgi ifodalarni (1) ga qo'yamiz hamda ketma–ket almashtirishlar bajaramiz:

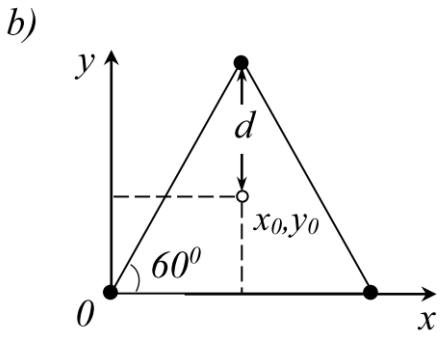
$$y_0 = \frac{(a/2)a^2 + (a + b/2\sqrt{3})b^2(\sqrt{3}/4)}{a^2 + b^2\sqrt{3}/4} = \frac{a}{2} + \frac{b^2(b + \sqrt{3}a)}{2(4a^2 + b^2\sqrt{3})}$$

Demak, masalaning ishchi formulasi:

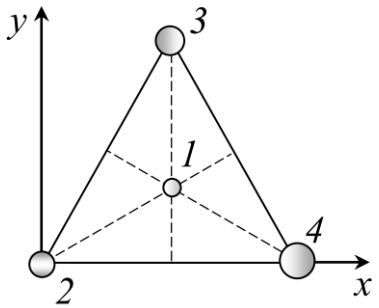
$$\boxed{y_0 = \frac{a}{2} + \frac{b^2(b + \sqrt{3}a)}{2(4a^2 + b^2\sqrt{3})}} \qquad (2)$$

Hisoblash natijasi: $y_0 = 19{,}2\ m$.

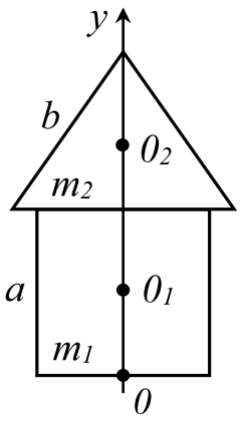




1-rasm



2-rasm



3-rasm

@olimpdep

Fan olimpiadalari bo‘yicha
iqtidorli o‘quvchilar bilan ishlash
DEPARTAMENTI